

A Structural Model of Rational Addiction

Archival English Reader's Translation of the June 2026 Manuscript (Repository v4.0)

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Archival note. This document is an abridged English reader's translation of the historical Chinese manuscript internally labeled **v6.2** and archived in this repository as **v4.0**. It is intended to make that research stage intelligible to non-Chinese readers. The original Chinese manuscript is the authoritative historical version. No attempt has been made here to retrofit later literature audits, later terminology, or later revisions into the June 2026 argument. Mathematical claims below reproduce the structure and main statements of that manuscript; some proofs, remarks, and intermediate derivations are summarized rather than translated line by line.

Abstract

This manuscript asks whether, within a rational-addiction framework, recovery can enter utility as an endogenous force that counteracts the dynamic reinforcement of addiction capital. Instantaneous utility is decomposed into reinforcement utility $F_0(c, A)$ and recovery utility $G(c, A, R)$, where $R = A - A^*$ is the gap between actual addiction stock and a target stock. Starting from axioms imposed on state-contingent choice, the manuscript states a representation result for G and constructs a kernel family whose recovery intensity may vary with the absolute addiction stock without requiring separability between consumption and stock.

Within this framework the manuscript derives five main results. First, the recovery component can reverse the sign of adjacent complementarity in the consumption–stock cross margin. Second, the steady-state equation is exactly affine in the exit-intensity parameter η , yielding a fold threshold for changes in the number of steady states. Third, when the target stock is required to be a Personal Equilibrium fixed point, a linear adjustment family fails to deliver such a fixed point under the stated parameter direction, while alternative nonlinear or target-dependent kernels can restore existence. Fourth, a delayed target-formation mechanism expands the state space and permits oscillatory local dynamics. Fifth, the numerical example locates a Neimark–Sacker bifurcation and computes a positive first Lyapunov coefficient, so the reported example is subcritical rather than the stable quasiperiodic cycle suggested by the initial motivation.

The manuscript also relates its construction to Becker–Murphy rational addiction, reference dependence, temptation/self-control representations, opponent-process ideas, and internal habit formation. Its concluding section explicitly leaves open dynamic self-correction, full endogenization of the target, identification of F_0 and G , and empirical calibration.

1 Motivation and historical position

The starting point is the reinforcement mechanism in Becker and Murphy (1988): current consumption raises a stock that increases the marginal utility of later consumption. With adjacent complementarity, a fully forward-looking optimizer can rationally select a high-consumption path. The June 2026 manuscript asks a different question from commitment-device or present-bias models: can a force associated with moving toward a lower target be represented *inside current utility itself*, and can the same fully rational agent evaluate both reinforcement and recovery?

The proposed decomposition is

$$U_t = F_0(c_t, A_t) + G(c_t, A_t, R_t), \quad R_t = A_t - A_t^*. \quad (1)$$

F_0 retains the standard reinforcement properties. G measures a target-relative recovery increment. The target A_t^* is treated as a state, not as a current control. In the baseline analysis it is fixed; later it follows an adaptive law of motion. This makes the target concept closer, in formal status, to an internal habit stock than to a commitment constraint.

The historical manuscript presented the following contribution map: an axiomatic representation and kernel construction; a “reinforcement–substitution reversal” in the cross derivative; a Personal Equilibrium analysis of the target stock; an exact fold boundary for steady-state multiplicity; and delayed target formation generating a two-dimensional local system capable of complex eigenvalues and a Neimark–Sacker bifurcation. Later versions of the broader project substantially revised the interpretation and novelty claims. This companion intentionally preserves the June 2026 framing rather than rewriting it from hindsight.

2 States and preferences

2.1 Addiction stock and recoverable stock

Let $A_t \geq 0$ denote the actual addiction stock at the beginning of period t . The standard accumulation equation is

$$A_{t+1} = (1 - \delta)A_t + c_t, \quad \delta \in (0, 1), \quad (2)$$

where $c_t \geq 0$ is current consumption. Let $A_t^* \geq 0$ be the target addiction stock and define

$$R_t \equiv A_t - A_t^*. \quad (3)$$

Thus $R_t > 0$ means that actual stock lies above the target and there is positive “recoverable” stock; $R_t < 0$ means actual stock lies below the target. In the baseline model $A_t^* \equiv A^*$ is exogenous and constant. Section 7 introduces a slow target-formation mechanism.

2.2 Axiomatic decomposition

For each state (A, R) , the primitive object is a conditional preference relation $\succeq_{(A,R)}$ over consumption $c \in \mathbb{R}_{\geq 0}$. The original manuscript imposes seven conditions:

- (P1) **Weak order:** completeness and transitivity at every state.
- (P2) **Continuity:** upper and lower contour sets in consumption are closed.
- (P3) **Addiction complementarity:** a numerical representation satisfies $F_c > 0$, $F_A > 0$, and $F_{cA} > 0$.
- (P4) **Reference independence at the target:** when $R = 0$, ranking depends only on (c, A) , giving a baseline function $F_0(c, A)$ with the standard rational-addiction properties.
- (P5) **Monotone crossing:** for $R > 0$, the recovery increment $G(c, A, R) = F(c, A, R) - F_0(c, A)$ is positive when $c < R$ and negative when $c > R$.
- (P6) **Diminishing marginal withdrawal gain:** $G_{cc} > 0$.
- (P7) **Diminishing marginal recovery space:** $G_{RR} < 0$.

Proposition 1 (Preference representation theorem). *If $\succeq_{(A,R)}$ satisfies the seven conditions above, the manuscript states that there is a continuous representation*

$$F(c, A, R) = F_0(c, A) + G(c, A, R), \quad (4)$$

with the following properties:

- (i) F_0 satisfies the baseline rational-addiction conditions;
- (ii) $G(c, A, 0) = 0$;
- (iii) $G(R, A, R) = 0$;
- (iv) $G_c < 0$;
- (v) $G_{cc} > 0$;
- (vi) $G_{RR} < 0$.

The proof in the historical manuscript proceeds by invoking continuous utility representation state by state, defining $F_0(c, A) = F(c, A, 0)$, setting $G = F - F_0$, using the crossing property and continuity to obtain $G(R, A, R) = 0$, and combining the crossing and convexity requirements to obtain the asserted sign of G_c . The document then notes that $-G$ has a saddle-type Hessian in (c, R) because $G_{cc}G_{RR} < 0$.

Remark 1 (Additive case). *If G does not depend on A , then $G_A \equiv 0$ and the model reduces to the additive form used in earlier drafts. Allowing $G_A \neq 0$ is intended to let two agents with the same gap R but different absolute addiction stocks receive different recovery increments.*

3 Operational recovery kernel

For direct utility, the manuscript assumes F_0 (denoted $\tilde{u}$ in some passages) satisfies

$$(F_0)_c > 0, \quad (F_0)_A > 0, \quad (F_0)_{cc} < 0, \quad (F_0)_{cA} > 0,$$

with sufficient smoothness. The positive cross derivative is the Becker–Murphy adjacent-complementarity channel.

The simplest recovery term is the linear benchmark

$$G(c_t, A_t, R_t) = \eta(R_t - c_t). \quad (5)$$

To avoid making that functional form central, the manuscript generalizes it to a kernel family:

$$G(c_t, A_t, R_t) = \int_{c_t}^{R_t} q(s, A_t) ds, \quad q(s, A) > 0, \quad q_s(s, A) < 0, \quad (6)$$

with

$$q(s, A; \eta) = \eta \tilde{q}(s, A), \quad \tilde{q} > 0, \quad \tilde{q}_s < 0. \quad (7)$$

Here $\eta > 0$ is an overall recovery-intensity parameter. The general kernel need not be multiplicatively separable in s and A . A special separable family is

$$\tilde{q}(s, A) = q_0(ks)\phi(A), \quad (8)$$

with $q_0(0) = 1$, $q'_0 < 0$, and $\phi(A) > 0$. Examples used later include $q_0(s) = e^{-ks}$ and $\phi(A) = 1 + \gamma A$.

The manuscript's kernel restrictions are: positivity; decreasing dependence on the consumption argument; recovery of the linear benchmark as a degenerate case; separability of the scale parameter η but not necessarily of (s, A) ; and a regularity/tail condition on q_s .

Proposition 2 (Automatic boundary condition). *For any G of the form (6),*

$$G(R_t, A_t, R_t) = \int_{R_t}^{R_t} q(s, A_t) ds = 0.$$

Hence this boundary condition is generated by the integral structure rather than imposed separately.

Proposition 3 (Dependence on absolute addiction stock). *For (6)–(7),*

$$G_A(c_t, A_t, R_t) = \eta \int_{c_t}^{R_t} \tilde{q}_A(s, A_t) ds.$$

Thus G_A is generally nonzero whenever the kernel responds to the absolute stock. Its sign is deliberately left unrestricted.

Proposition 4 (Reinforcement–substitution reversal). *For the composite flow utility $U(c, A, R) = F_0(c, A) + G(c, A, R)$,*

$$U_{cA}(c, A) = (F_0)_{cA}(c, A) - \eta \tilde{q}_A(c, A), \quad (9)$$

and the expression is independent of R for this kernel family. Consequently:

(i) *if $\tilde{q}_A \equiv 0$, then $U_{cA} = (F_0)_{cA} > 0$ and adjacent complementarity is preserved;*

(ii) *at a fixed (c, A) , defining $\tilde{q}_A^* = (F_0)_{cA}/\eta$, one has*

$$\tilde{q}_A(c, A) > \tilde{q}_A^*(c, A) \iff U_{cA}(c, A) < 0;$$

(iii) *therefore reversal requires $\tilde{q}_A > 0$ at the point in question.*

A further extension in the original manuscript allows the kernel itself to depend on the target gap:

$$G(c, A, R) = \int_c^R q(s, A, R) ds. \quad (10)$$

Then

$$G_{cR}(c, A, R) = -q_R(c, A, R)$$

need not vanish, and

$$G_R = q(R, A, R) + \int_c^R q_R(s, A, R) ds, \quad G_A = \int_c^R q_A(s, A, R) ds.$$

This extension is later used as one route to restore Personal Equilibrium existence.

4 Dynamic programming and optimality

The forward-looking agent solves

$$V(A_t; A_t^*) = \max_{c_t} \{F_0(c_t, A_t) + G(c_t, A_t, R_t) + \beta_d V(A_{t+1}; A_t^*)\}, \quad (11)$$

subject to (2), where $\beta_d \in (0, 1)$.

For an interior choice the first-order condition is

$$(F_0)_c(c_t, A_t) - q(c_t, A_t) + \beta_d V'(A_{t+1}; A_t^*) = 0. \quad (12)$$

The envelope equation must account for the fact that A_t affects recovery both directly and through $R_t = A_t - A_t^*$; in the notation of the manuscript this adds the combination $G_A + G_R$ to the usual stock derivative.

The draft emphasized that the recovery component and continuation value are intended as distinct incentive channels. In the limiting case $\beta_d = 0$, G_c can still change the optimal current choice; in the limiting case $\eta \rightarrow 0$, the continuation value can still change the current choice. This observation was used in the historical draft to argue against interpreting G and $\beta_d V$ as mechanically duplicated terms.

5 Steady states and the fold boundary

At a stationary solution let

$$A_t = A_{t+1} = \bar{A} = \frac{c^*}{\delta}, \quad \bar{R} = \frac{c^*}{\delta} - A^*.$$

Write

$$r \equiv 1 - \beta_d(1 - \delta).$$

Eliminating the shadow value using the first-order and envelope conditions produces a scalar steady-state equation

$$\Phi(c^*; \eta) = 0. \quad (13)$$

The key structural observation of the June draft is that, because $q = \eta \tilde{q}$, this equation is exactly affine in η .

Proposition 5 (Exact affine structure in η). *The steady-state equation can be written*

$$\Phi(c^*; \eta) = \Phi_0(c^*) + \eta \Xi(c^*), \quad (14)$$

where

$$\begin{aligned} \Phi_0(c^*) &= (F_0)_c(c^*, \bar{A}) + \frac{\beta_d}{r} (F_0)_A(c^*, \bar{A}), \\ \Xi(c^*) &= -\tilde{q}(c^*, \bar{A}) + \frac{\beta_d}{r} \left[\int_{c^*}^{\bar{R}} \tilde{q}_A(s, \bar{A}) ds + \tilde{q}(\bar{R}, \bar{A}) \right], \end{aligned}$$

with $\bar{A} = c^*/\delta$ and $\bar{R} = c^*/\delta - A^*$. Thus $\partial\Phi/\partial\eta = \Xi(c^*)$ exactly, not merely as a local approximation.

Proposition 6 (Sufficient multiplicity condition). *Under the regularity conditions of the manuscript, if adjacent complementarity is strong enough to make Φ fall over an interval, the tail of Φ is nonnegative, and η is large enough to make the minimum of Φ negative, then the steady-state equation has at least two positive roots. The draft interprets these as low- and high-consumption steady states separated by an unstable threshold.*

Because of (14), define

$$H(c^*) = -\frac{\Phi_0(c^*)}{\Xi(c^*)} \quad (15)$$

where $\Xi \neq 0$.

Proposition 7 (Exact fold characterization). *Suppose $\Xi(c^\dagger) \neq 0$ and $\Xi'(c^\dagger) \neq 0$. There is a positive $\eta^\dagger$ such that*

$$\Phi(c^\dagger; \eta^\dagger) = 0, \quad \frac{d\Phi}{dc^*}(c^\dagger; \eta^\dagger) = 0$$

if and only if $c^\dagger$ is a critical point of H and $\eta^\dagger = H(c^\dagger)$. Away from such degenerate roots, simple roots persist locally, so the number of positive roots can change only at these critical values.

5.1 Illustrative numerical example

The numerical example is explicitly illustrative rather than calibrated. It uses

$$F_0(c, A) = \ln c + \alpha c A, \quad q_0(s) = e^{-ks}, \quad \phi(A) = 1 + \gamma A,$$

with the following parameter values.

Table 1: Illustrative parameter values from the June 2026 manuscript

Parameter	Symbol	Value	Role
Depreciation rate	δ	0.1	stock law
Discount factor	β_d	0.95	intertemporal valuation
Adjacent-complementarity strength	α	1.0	$F_0 = \ln c + \alpha c A$
Withdrawal response rate	k	1.0	$q_0(s) = e^{-ks}$
Stock modulation	γ	0.5	$\phi(A) = 1 + \gamma A$
Target stock	A^*	2.0	exogenous in this example
Derived stock-discount factor	r	≈ 0.145	$1 - \beta_d(1 - \delta)$

Solving $H'(c^*) = 0$ numerically gives

$$c^\dagger \approx 1.0190, \quad \eta_{\text{fold}}^* = H(c^\dagger) \approx 17.716.$$

For $\eta = 10$ the displayed region has no interior root; at $\eta = \eta_{\text{fold}}^*$ the graph is tangent to zero; and for $\eta = 25$ it has two roots near 0.735 and 1.686.

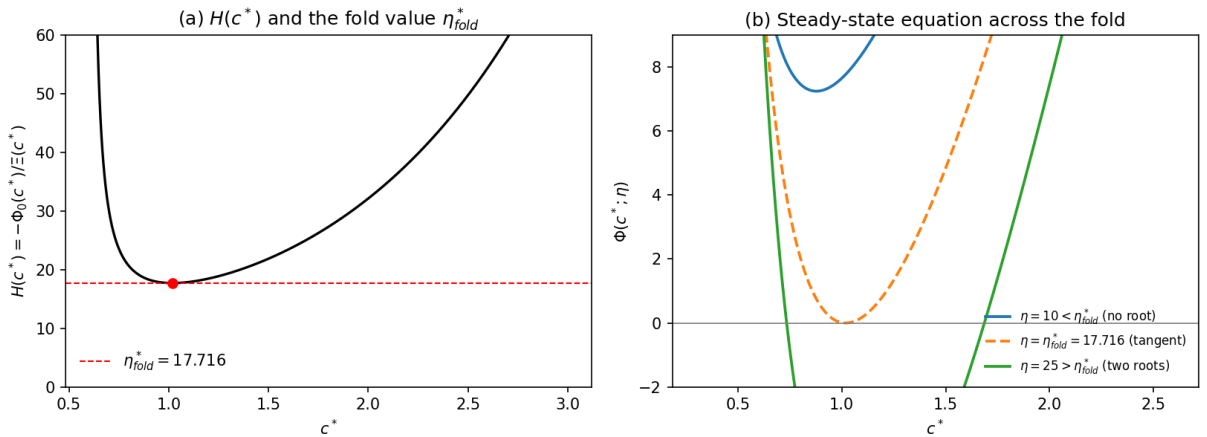


Figure 1: Historical fold illustration. Panel (a) shows $H(c^*) = -\Phi_0(c^*)/\Xi(c^*)$ and its local minimum $\eta_{\text{fold}}^* \approx 17.716$ at $c^\dagger \approx 1.019$. Panel (b) shows the steady-state equation before, at, and above the fold threshold: no root for $\eta = 10$, tangency at the threshold, and two positive roots for $\eta = 25$.

6 Target stock as a Personal Equilibrium

The manuscript next asks whether the target can be interpreted as a rational-expectations fixed point: the target stock equals the steady-state stock generated when that target is maintained. If A^{*PE} is such a Personal Equilibrium, then $c^{*PE} = \delta A^{*PE}$ and the recoverable gap is zero at the fixed point.

Proposition 8 (Algebraic characterization of Personal Equilibrium). *A^{*PE} is a Personal Equilibrium if and only if $c^{*PE} = \delta A^{*PE}$ is a positive root of the steady-state equation evaluated at $\bar{R} = 0$. In the notation of the original draft,*

$$\Phi^{PE}(c^*) = (F_0)_c(c^*, \bar{A}) - q(c^*, \bar{A}) + \frac{\beta_d}{r} \left[(F_0)_A(c^*, \bar{A}) + \eta \int_{c^*}^0 \tilde{q}_A(s, \bar{A}) ds + \eta \tilde{q}(0, \bar{A}) \right] = 0,$$

where $\bar{A} = c^*/\delta$.

Proposition 9 (Nonexistence under linear modulation). *In the separable family with $\phi(A) = 1 + \gamma A$, if $\beta_d/r > 1$, then for every $\gamma \geq 0$ the manuscript obtains $\Phi^{PE}(c^*) > 0$ for all $c^* > 0$. Hence no Personal Equilibrium exists in this linear-modulation case.*

The draft emphasizes a tension: the cross-derivative reversal in Proposition 4 naturally points toward $\phi'(A) > 0$, while the simple linear family with that direction fails the Personal Equilibrium fixed-point requirement.

Two routes are then developed to show that the tension belongs to the particular tractable family rather than to the broad idea itself.

Proposition 10 (Existence under exponential modulation). *Let $\phi(A) = \phi_0 e^{\kappa A}$ with $\phi_0, \kappa > 0$, let $q_0(s) = e^{-ks}$, assume $\kappa > k$, and suppose $\beta_d/r > 1$. Then the manuscript states that a Personal Equilibrium exists for every $\eta > 0$.*

Proposition 11 (Compatibility with reversal). *Under the preceding assumptions, if the resulting Personal Equilibrium stock is sufficiently large, the reinforcement–substitution reversal condition can also hold at that fixed point.*

The cost of this route, noted by the manuscript itself, is that the exponential modulation can conflict with growth restrictions implicit in the earlier fold analysis.

Proposition 12 (Existence with an R -dependent kernel). *Alternatively, keep $\phi(A) = 1 + \gamma A$ with $\gamma > 0$ but allow*

$$q(s, A, R) = \eta q_0(ks) \phi(A) \chi(R),$$

with $\chi(0) = \chi_0 > 0$ and $\chi'(0) = \chi'_0$. If

$$\chi'_0 > k\chi_0,$$

then there is a threshold

$$\eta_0 = \frac{\alpha(1 + \delta\beta_d/r)}{\gamma(\beta_d/r)(\chi'_0/k - \chi_0)}$$

such that a Personal Equilibrium exists for all $\eta > \eta_0$.

The June draft therefore treats exponential stock modulation and target-gap dependence as two distinct ways to repair the linear-family fixed-point failure.

7 Delayed target formation and local dynamics

The last major extension replaces the fixed target with an adaptive state. The manuscript first motivates a lagged rule and then analyzes the smooth two-dimensional specification

$$A_{t+1}^* = (1 - \rho)A_t^* + \rho w(A_t), \quad \rho \in (0, 1). \quad (16)$$

The actual stock continues to satisfy

$$A_{t+1} = (1 - \delta)A_t + c^*(A_t, A_t^*).$$

The agent chooses only current consumption. Both stocks evolve according to given transition rules; in that formal sense the target is treated as a state rather than as a control.

Let $(\bar{A}, \bar{A}^*)$ be a steady state and define the local policy sensitivities

$$c_A = \left. \frac{\partial c^*}{\partial A_t} \right|_{ss}, \quad c_{A^*} = \left. \frac{\partial c^*}{\partial A_t^*} \right|_{ss}.$$

The first can be expressed from the implicit first-order condition as

$$c_A = - \left. \frac{(F_0)_{cA} - q_A + \beta_d(1 - \delta)V''}{(F_0)_{cc} - q_c + \beta_d V''} \right|_{ss}, \quad (17)$$

while c_{A^*} depends on a cross derivative of the value function and is treated as a local policy characteristic in the general analysis.

Linearization in $x_t = A_t - \bar{A}$ and $y_t = A_t^* - \bar{A}^*$ gives

$$\begin{pmatrix} x_{t+1} \\ y_{t+1} \end{pmatrix} = J \begin{pmatrix} x_t \\ y_t \end{pmatrix}, \quad J = \begin{pmatrix} 1 - \delta + c_A & c_{A^*} \\ \rho w_A & 1 - \rho \end{pmatrix}, \quad (18)$$

where $w_A = w'(\bar{A})$. The characteristic polynomial is

$$\lambda^2 - T\lambda + D = 0, \quad (19)$$

with

$$T = 2 - \delta - \rho + c_A, \quad D = (1 - \delta + c_A)(1 - \rho) - c_{A^*}\rho w_A. \quad (20)$$

Proposition 13 (Oscillatory adjustment). *If*

$$(2 - \delta - \rho + c_A)^2 < 4[(1 - \delta + c_A)(1 - \rho) - c_{A^*}\rho w_A],$$

then the eigenvalues are a complex-conjugate pair. If $D < 1$ their modulus is below one and oscillations decay; if $D = 1$ the linearized amplitude is constant; if $D > 1$ the amplitude grows and the steady state is locally unstable.

Proposition 14 (Neimark–Sacker condition). *Use ρ as a bifurcation parameter and let ρ^* satisfy $D(\rho^*) = 1$. The June manuscript states a Neimark–Sacker bifurcation when the following hold at ρ^* :*

(NS1) $T^2 < 4$ (complex roots);

(NS2) $dD/d\rho = -(1 - \delta + c_A) - c_{A^*}w_A \neq 0$ (transversality, in the local formulation used in the draft);

(NS3) $T \neq -1$ and $T \neq 0$ (nonresonance as stated there);

(NS4) the first Lyapunov coefficient of the nonlinear system satisfies $l_1 \neq 0$.

The sign convention used in the manuscript classifies $l_1 < 0$ as supercritical with a stable invariant curve, and $l_1 > 0$ as subcritical with an unstable invariant curve.

7.1 Historical numerical bifurcation example

For the numerical illustration, the draft continues with

$$F_0(c, A) = \ln c + \alpha c A, \quad q_0(s) = e^{-ks}, \quad \phi(A) = 1 + \gamma A,$$

and chooses a linear target rule

$$w(A) = \omega A, \quad \omega = 0.483.$$

The selected steady state is reported as

$$\bar{c} \approx 1.21495, \quad \bar{A} \approx 12.150, \quad \bar{A}^* \approx 5.868, \quad \bar{R} \approx 6.281.$$

Solving for the local sensitivities and locating $D(\rho) = 1$ yields

$$\rho^* \approx 0.029261,$$

with

$$T(\rho^*) \approx 1.999143, \quad D(\rho^*) \approx 1, \quad T^2 - 4D \approx -0.003426 < 0.$$

The eigenvalues are reported as approximately

$$0.99957 \pm 0.02926i,$$

with unit modulus. The numerical derivative $dD/d\rho \approx -1.990$ is nonzero. The implied linearized rotation angle is about 0.02927 radians, corresponding to roughly 215 periods; the manuscript explicitly notes that this is not a calibration to an observed relapse frequency.

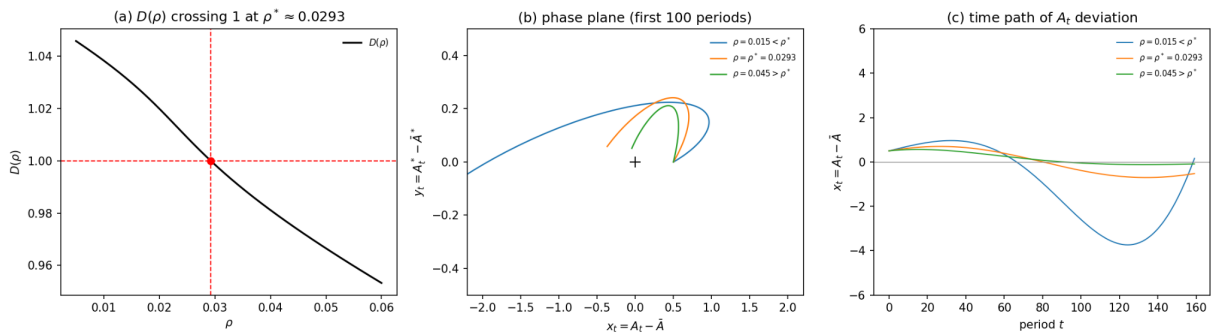


Figure 2: Historical Neimark-Sacker illustration. Panel (a) shows $D(\rho)$ crossing one at $\rho^* \approx 0.0293$. Panel (b) shows the first 100 periods in the phase plane for values below, at, and above the critical ρ . Panel (c) shows the corresponding path of $A_t - \bar{A}$. In this numerical configuration D decreases with ρ , so lower ρ gives an expanding spiral and higher ρ a contracting spiral.

To check the nonlinear nondegeneracy condition, the manuscript recursively solves second- and third-order local derivatives of the policy and shadow value. It reports

$$c_{AA} = -0.27897, \quad c_{A,A^*} = 0.59029, \quad c_{A^*A^*} = -1.85996,$$

$$c_{AAA} = 3.31395, \quad c_{AA,A^*} = -10.74608, \quad c_{A,A^*A^*} = 34.56228, \quad c_{A^*A^*A^*} = -145.36327.$$

Using the normal-form calculation described in the Chinese manuscript gives

$$l_1 \approx 0.4906 > 0. \tag{21}$$

The historical draft therefore concludes that this particular numerical Neimark–Sacker bifurcation is *subcritical*. It explicitly acknowledges that this example does not generate the stable quasiperiodic relapse–cessation cycle envisioned in the motivating narrative; a stable invariant curve would require the opposite sign in the convention used there.

8 What the June 2026 manuscript left open

The conclusion of the original manuscript is unusually explicit about its own boundaries. Four open issues are central.

1. Dynamic self-correction. The cross derivative in (9) can reverse adjacent complementarity, but the manuscript does not prove that the optimal policy derivative c_A (or the reduced one-dimensional derivative in a degenerate case) must be negative. The result is therefore a flow-level countervailing incentive, not a general theorem that the equilibrium path must converge toward the target.

2. Full endogenization of the target. The target can be treated as an adaptive state without requiring a Personal Equilibrium. The Personal Equilibrium exercise is only one particular rational-expectations interpretation. The linear separable family fails that fixed-point test; two modifications restore existence, but each changes other parts of the tractable structure. A complete unification is left unfinished.

3. Identification. Because F_0 and G enter the Euler equation additively, the draft does not establish that consumption–stock path data can separately identify the baseline reinforcement component and the recovery kernel parameters. The delayed two-dimensional dynamics are suggested as a possible source of testable distinctions, but no econometric identification result is supplied.

4. Calibration. Both numerical exercises are illustrative. The fold example checks the algebraic threshold; the bifurcation example verifies a full set of local conditions for one parameter configuration. The manuscript does not calibrate the model to clinical or consumption data, and the reported positive Lyapunov coefficient means the chosen example is not a stable quasiperiodic cycle.

9 Archival summary

At this stage of the research program, the conceptual object was an endogenous recovery term evaluated by the same forward-looking agent who values addictive consumption. The mathematical program moved from a direct functional form to a kernel family, then to state-dependent recovery, target fixed points, exact steady-state fold thresholds, and a delayed two-state dynamical system. The June 2026 manuscript’s strongest internal lesson was already visible in its conclusion: reversing a local cross margin is not the same as proving global or even local dynamic self-correction, and

the numerical Neimark–Sacker example can exist while still failing to deliver the stable relapse cycle that motivated the project.

This English companion is preserved to make that historical stage readable. Later versions of the project changed the primitives, terminology, and interpretation substantially; those later versions should be consulted for the author’s subsequent position.

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